

Pop Quiz*Two-Step Equations* **ANSWER KEY**

DIRECTIONS. Solve each equation. Show the step that undoes the addition or subtraction and the step that undoes the multiplication.

1. $3x + 7 = 22$

$$3x + 7 - 7 = 22 - 7$$

$$3x = 15$$

$$x = 5$$

2. $5y - 13 = 22$

$$5y - 13 + 13 = 22 + 13$$

$$5y = 35$$

$$y = 7$$

3. $-2x + 9 = 1$

$$-2x + 9 - 9 = 1 - 9$$

$$-2x = -8$$

$$x = 4$$

4. $23 = 3m - 4$

$$23 + 4 = 3m - 4 + 4$$

$$27 = 3m$$

$$m = 9$$

5. $6(x + 2) = 48$

$$6(x + 2) \div 6 = 48 \div 6$$

$$x + 2 = 8$$

$$x = 6$$

6. $0.5p + 3 = 10$

$$0.5p + 3 - 3 = 10 - 3$$

$$0.5p = 7$$

$$p = 14$$

7. A gym charges a one-time sign-up fee of \$20, then \$15 for each month. Marcus paid \$95 in all. (2 points)

a. Write an equation. Let m be the number of months.

$$20 + 15m = 95$$

b. Solve it. How many months?

$$20 + 15m - 20 = 95 - 20$$

$$15m = 75$$

$$m = 5$$

Marcus paid for 5 months.

$$\text{Check: } 20 + 15(5) = 20 + 75 = 95 \checkmark$$

GRADING. One point per lettered part, and every part is right or it is not. Questions 1–6 are one point each; question 7 has two parts, a and b, and each is one point. Eight points in all. There is no partial credit: a correct answer reached without the shown step still earns its point, and the working is what you circulate for, not what you weigh.

WHAT TO WATCH FOR. Question 3: $-2x = -8$ gives $x = 4$, not -4 . Question 4 puts the variable on the right; a student who writes $23 - 4$ has subtracted from the wrong side. Question 5 divides by 6 first — distributing to $6x + 12 = 48$ is equally correct and earns the point. Question 6 is the one where dividing by a decimal trips people: $7 \div 0.5 = 14$, not 3.5.